Fiscal Year:	FY 2020	**Task Last Updated:**	FY 01/14/2020
PI Name:	Wolverton, Scot Ph.D.		
Project Title:	Characterizing Plant Gravity Perception Systems		
Division Name:	Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Cellular and molecular biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Plant Biology		
Space Biology Cross-Element Discipline:	(1) Developmental Biology		
Space Biology Special Category:	(1) Bioregenerative Life Support		
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Zip Code:	43015-2333	**Congressional District:**	12
Comments:			
Project Type:	Flight,Ground	**Solicitation / Funding Source:**	2014 Space Biology Flight (ILSRA) NNH14ZTT002N
Start Date:	03/16/2015	**End Date:**	03/15/2021
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	4
No. of Bachelor's Candidates:	6	**Monitoring Center:**	NASA KSC
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Flight Program:	ISS		
Flight Assignment:	ISS NOTE: Extended to 3/15/2021 per NSSC information (Ed., 12/16/19) NOTE: Extended to 3/15/2020 per NSSC information (Ed., 12/17/18) NOTE: Extended to 3/15/2019 per NSSC information (Ed., 12/18/17)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NNX15AG55G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Plant growth is characterized by plasticity, allowing environmental cues like light and gravity to play a significant role in determining the overall form of the plant body. Statolith sedimentation in specialized cells represents one way that plants perceive gravity. Plants lacking functional statoliths show reduced but consistent gravity responses, suggesting the presence of a non-statolith sensory system. This proposal seeks to investigate the gravity perception machinery in roots of several genotypes of the higher plant Arabidopsis thaliana by taking advantage of the unique microgravity environment of Earth orbit on the International Space Station (ISS). One thrust of the work described here is to characterize the physiological properties of the putative non-statolith sensory system, including the threshold acceleration force required to activate it, and its capacity to modulate downstream growth regulation machinery. The second thrust is to determine whether the two gravity sensory systems control a single or multiple output systems. Addressing this question requires the microgravity environment in order to quantify the kinetics of gravitropism in roots having or lacking functional statoliths under fractional g conditions. The overall outcome of the work proposed here will be an improved understanding of how higher plants perceive gravity and transduce this physical cue into directional growth, a fundamental requirement for improving plants for production on Earth and in long-term spaceflight missions.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The overall outcome of the work proposed here will be an improved understanding of how higher plants perceive gravity and transduce this physical cue into directional growth, a fundamental requirement for improving plants for production on Earth and in long-term spaceflight missions. In particular, the differences in growth responses and gene regulation between wild-type and starchless mutants will help elucidate the contributions of statolith-based sensing to plant gravity responses.
Task Progress:	In this, the fourth year of funding (March 2019 — March 2020), we completed the analysis of growth responses from the spaceflight experiments; we conducted 1 g ground experiments in flight-like hardware; we completed a pilot study of gene expression differences between wild-type and starchless mutants (pgm-1); we used the results of the pilot study to conduct a mutant screen. Each of these tasks is discussed in greater detail below. The Plant Gravity Perception (PGP) payload was launched on SpaceX CRS13 in December of 2017 and operations were carried out January 26 - April 12, 2018. The duration of the flight experiments was increased substantially due to the unpredictable behavior of the European Modular Cultivation System (EMCS) facility on the ISS, which had problems powering the hardware throughout all four runs. Despite this difficulty the main objectives of PGP were accomplished with a minimum of science loss thanks to the tremendous effort of all team members. Germination rates for both wild-type and starchless mutants was approximately 84% for the spaceflight experiment, yielding more than sufficient quantities of seedlings for both image analysis of the responses to fractional gravity and light as well as for isolation of RNA for RNAseq analysis. Seedlings on orbit demonstrated clear and consistent responses to unilateral blue light stimulation in microgravity conditions (no fractional g treatment), with roots orienting away from the direction of light stimulation as expected. The response of seedling roots to the ensuing fractional gravity treatment showed a strong dependence upon the genotype of the plant, with wild-type roots having a much greater sensitivity to fractional g treatments than starchless mutants. This was also predicted, but the magnitude of difference was unknown and will inform our interpretation of the molecular-level differences we anticipate from the RNAseq data. In addition to the flight experiments, we also successfully carried out a ground-based pilot RNAseq experiment in preparation for processing the flight tissue samples, which returned in May 2018. We designed an experiment using wild-type and pgm-1 starchless mutants in which we collected tissue for processing from vertically-growing seedlings and seedlings receiving a 90 degree reorientation 10 min prior to freezing and tissue harvesting. Following freezing, all seedlings were dissected into root tips, root bases, hypocotyls, and cotyledons and RNA isolation was carried out on 4-7 seedling tissues per type with 4 replicates of each treatment and genotype. RNA samples were used for cDNA synthesis and library preparation using a low-input cDNA library kit, and libraries were sequenced on an Illumina NovaSeq 6000. We have completed differential gene expression analysis now, including a contrast analysis, which resulted in the identification of more than 100 genes with contrasting expression patterns across treatments and genotypes. This ground-based data set will form a critical resource for comparison against our flight data set as well as providing a rich resource of candidate genes to be tested for a role in gravity perception or signal transduction. As a first step toward testing the function of these genes in the gravity response pathway, we obtained T-DNA insertional mutants in 35 of these genes showing contrasting expression and carried out a series of phenotyping experiments. In addition to the standard root gravitropism assay, we developed a new assay based on our ROTATO image analysis and feedback rotation system that stimulates the root by 90 deg, then continues to rotate the plate at a constant rate of rotation. This technique requires the root to continue expressing differential growth, something that wild-type roots do with ease, but roots defective in gravity perception or response fail to maintain. This combination of assays has identified 10 of the 35 mutants tested thus far to show some degree of gravity-related phenotype. We believe this approach holds promise to identify and map more of the gravity response machinery, and we look forward to building on these data and those that emerge from the differential expression data from seedlings exposed to fractional amounts of g.
Bibliography Type:	Description: (Last Updated: 05/04/2026)
Articles in Peer-reviewed Journals	Kiss JZ, Wolverton C, Wyatt SE, Hasenstein KH, van Loon JJWA. "Comparison of microgravity analogs to spaceflight in studies of plant growth and development." Front Plant Sci. 2019 Dec 6;10:1577. Review. https://doi.org/10.3389/fpls.2019.01577 ; PubMed PMID: 31867033; PubMed Central PMCID: PMC6908503 , Dec-2019